

8 *Determining Process and Measurement Systems Capability*

CHAPTER OUTLINE

- | | | | |
|-------|---|-------|--|
| 8.1 | INTRODUCTION | 8.7.2 | The Analysis of Variance Method |
| 8.2 | PROCESS CAPABILITY ANALYSIS USING A HISTOGRAM OR A PROBABILITY PLOT | 8.7.3 | Confidence Intervals in Gauge R & R Studies |
| 8.2.1 | Using the Histogram | 8.7.4 | False Defectives and Passed Defectives |
| 8.2.2 | Probability Plotting | 8.7.5 | Attribute Gauge Capability |
| 8.3 | PROCESS CAPABILITY RATIOS | 8.7.6 | Comparing Customer and Supplier Measurement Systems |
| 8.3.1 | Use and Interpretation of C_p | 8.8 | SETTING SPECIFICATION LIMITS ON DISCRETE COMPONENTS |
| 8.3.2 | Process Capability Ratio for an Off-Center Process | 8.8.1 | Linear Combinations |
| 8.3.3 | Normality and the Process Capability Ratio | 8.8.2 | Nonlinear Combinations |
| 8.3.4 | More about Process Centering | 8.9 | ESTIMATING THE NATURAL TOLERANCE LIMITS OF A PROCESS |
| 8.3.5 | Confidence Intervals and Tests on Process Capability Ratios | 8.9.1 | Tolerance Limits Based on the Normal Distribution |
| 8.4 | PROCESS CAPABILITY ANALYSIS USING A CONTROL CHART | 8.9.2 | Nonparametric Tolerance Limits |
| 8.5 | PROCESS CAPABILITY ANALYSIS USING DESIGNED EXPERIMENTS | | |
| 8.6 | PROCESS CAPABILITY ANALYSIS WITH ATTRIBUTE DATA | | |
| 8.7 | GAUGE AND MEASUREMENT SYSTEM CAPABILITY STUDIES | | |
| 8.7.1 | Basic Concepts of Gauge Capability | | |
- Supplemental Material for Chapter 8**
- | | |
|------|--|
| S8.1 | Fixed Versus Random Factors in the Analysis of Variance |
| S8.2 | More About Analysis of Variance Methods for Measurement Systems Capability Studies |

Learning Objectives

1. Investigate and analyze process capability using control charts, histograms, and probability plots
2. Understand the difference between process capability and process potential
3. Calculate and properly interpret process capability ratios
4. Understand the role of the normal distribution in interpreting most process capability ratios
5. Calculate confidence intervals on process capability ratios
6. Know how to conduct and analyze a measurement systems capability (or gauge R & R) experiment
7. Know how to estimate the components of variability in a measurement system
8. Know how to set specifications on components in a system involving interaction components to ensure that overall system requirements are met
9. Estimate the natural limits of a process from a sample of data from that process

Process Capability

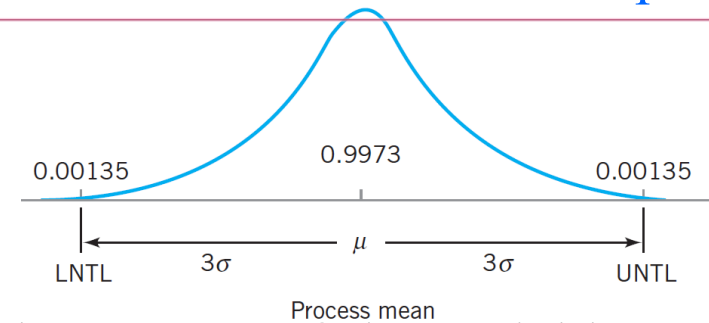
- Statistical techniques can be helpful throughout the product cycle, including development activities prior to manufacturing, in quantifying process variability, in analyzing this variability relative to product requirements or specifications, and in assisting development and manufacturing in eliminating or greatly reducing this variability. This general activity is called process capability analysis.
- Process capability refers to the uniformity of the process. Obviously, the variability of critical-to-quality characteristics in the process is a measure of the uniformity of output.
- There are two ways to think of this variability:
 1. The natural or inherent variability in a critical-to-quality characteristic at a specified time—that is, “instantaneous” variability
 2. The variability in a critical-to-quality characteristic over time

Process Capability

- It is customary to take the Six Sigma spread in the distribution of the product quality characteristic as a measure of process capability.
- The figure shows a process for which the quality characteristic has a normal distribution with mean μ and standard deviation σ . The upper and lower natural tolerance limits of the process fall at $\mu + 3\sigma$ and $\mu - 3\sigma$, respectively—that is,

$$\text{UNTL} = \mu + 3\sigma$$

$$\text{LNTL} = \mu - 3\sigma$$



- For a normal distribution, the natural tolerance limits include 99.73% of the variable, or put another way, only 0.27% of the process output will fall outside the natural tolerance limits.
- Two points should be remembered:
 - 0.27% outside the natural tolerances sounds small, but this corresponds to 2,700 nonconforming parts per million.
 - If the distribution of process output is non-normal, then the percentage of output falling outside $\pm 3s$ may differ considerably from 0.27%.

Process Capability

- We define **process capability analysis** as a formal study to estimate **process capability**. The estimate of process capability may be in the form of a probability distribution having a specified shape, center (mean), and spread (standard deviation).
- For example, we may determine that the process output is normally distributed with mean $\mu = 1$ cm and standard deviation $\sigma = 0.001$ cm.
- In this sense, a process capability analysis may be performed without regard to specifications on the quality characteristic.
- Alternatively, **we may express process capability as a percentage outside of specifications**.
However, specifications are not necessary to process capability analysis.

Process Capability

- Uses of process capability data:
 1. Predicting how well the process will hold the tolerances
 2. Assisting product developers/designers in selecting or modifying a process
 3. Assisting in establishing an interval between sampling for process monitoring
 4. Specifying performance requirements for new equipment
 5. Selecting between competing suppliers and other aspects of supply chain management
 6. Planning the sequence of production processes when there is an interactive effect of processes on tolerances
 7. Reducing the variability in a process

8.2 Process Capability Analysis Using a Histogram or a Probability Plot

Histogram

- The histogram can be helpful in estimating process capability.
- At least 100 or more observations should be available for the histogram (or the stem-and-leaf plot) to be moderately stable so that a reasonably reliable estimate of process capability may be obtained.
- The histogram, along with the sample average $\bar{x}$ and sample standard deviation s , provides information about process capability.

EXAMPLE 8.1 Estimating Process Capability with a Histogram

Figure 8.2 presents a histogram of the bursting strength of 100 glass containers. The data are shown in Table 8.1. What is the capability of the process?

SOLUTION

Analysis of the 100 observations gives

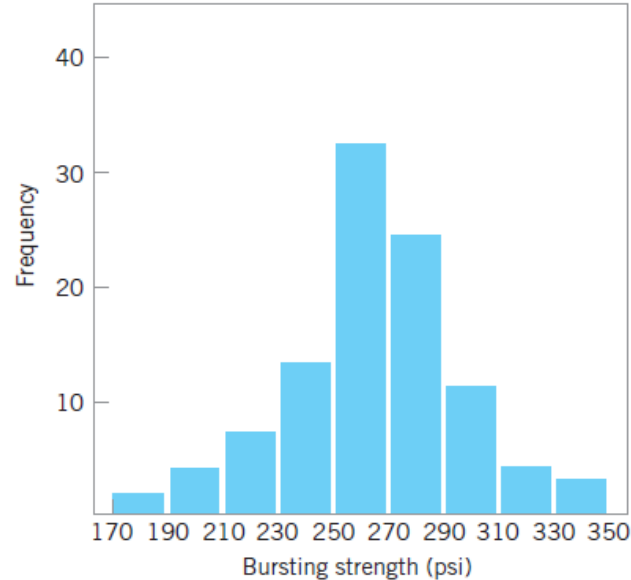
$$\bar{x} = 264.06 \quad s = 32.02$$

Consequently, the process capability would be estimated as

$$\bar{x} \pm 3s$$

or

$$264.06 \pm 3(32.02) \approx 264 \pm 96 \text{ psi}$$



■ **FIGURE 8.2** Histogram for the bursting-strength data.

Furthermore, the shape of the histogram implies that the distribution of bursting strength is approximately normal. Thus, we can estimate that approximately 99.73% of the bottles manufactured by this process will burst between 168 and 360 psi. Note that we can estimate process capability *independently of the specifications on bursting strength*.

■ **TABLE 8.1**

Bursting Strengths for 100 Glass Containers

265	197	346	280	265	200	221	265	261	278
205	286	317	242	254	235	176	262	248	250
263	274	242	260	281	246	248	271	260	265
307	243	258	321	294	328	263	245	274	270
220	231	276	228	223	296	231	301	337	298
268	267	300	250	260	276	334	280	250	257
260	281	208	299	308	264	280	274	278	210
234	265	187	258	235	269	265	253	254	280
299	214	264	267	283	235	272	287	274	269
215	318	271	293	277	290	283	258	275	251

8.2 Process Capability Analysis Using a Histogram or a Probability Plot

Probability Plotting

- Probability plotting is an alternative to the histogram that can be used to determine the shape, center, and spread of the distribution.
- It has the advantage that it is unnecessary to divide the range of the variable into class intervals, and it often produces reasonable results for moderately small samples (which the histogram will not).
- Generally, a probability plot is a graph of the ranked data versus the sample cumulative frequency on special paper with a vertical scale chosen so that the cumulative distribution of the assumed type is a straight line.
- An obvious disadvantage of probability plotting is that it is not an objective procedure. It is possible for two analysts to arrive at different conclusions using the same data.

8.3 Process Capability Ratios

- In a practical application, the process standard deviation σ is almost always unknown and must be replaced by an estimate $\hat{\sigma}$.
- To estimate σ we typically use either the sample standard deviation s or $\bar{R}/d_2$ (when variables control charts are used in the capability study). This results in an estimate of C_p —say,

$$\bullet \hat{C}_p = \frac{USL - LSL}{6\hat{\sigma}}$$

- This PCR C_p has a useful practical interpretation—namely

نسبة الانتاج
بالمعايير

$$\bullet P = \left(\frac{1}{\hat{C}_p}\right)100$$

is the percentage of the specification band used up by the process.

One-Sided PCR

- For one-sided specifications, one-sided process-capability ratios are used. One-sided PCRs are defined as follows.

$$C_{pu} = \frac{USL - \mu}{3\sigma} \quad (\text{upper specification only})$$

$$C_{pl} = \frac{\mu - LSL}{3\sigma} \quad (\text{lower specification only})$$

Estimates $\hat{C}_{pu}$ and $\hat{C}_{pl}$ would be obtained by replacing μ and σ by estimates $\hat{\mu}$ and $\hat{\sigma}$, respectively.

* EXAMPLE 8.2

- Construct a one-sided process-capability ratio for the container bursting-strength data in Example 8.1. Suppose that the lower specification limit on bursting strength is 200 psi.

SOLUTION

$$\bar{x} = 264 \text{ and } s = 32 \implies C_{pl} = \frac{264 - 200}{3(32)} = 0.67 < 1$$

Defects Per million

$$Z = \frac{LSL - \mu}{\sigma}$$

- The fraction of defective containers produced by this process is estimated by finding the area to the left of $Z = \frac{200 - 264}{(32)} = -2$. Thus, The estimated fallout is about 2.28% defective, or about 22,800 nonconforming containers per million.

Interpretation of the PCR

- The process capability ratio is a measure of the ability of the process to manufacture product that meets the specifications.
- Table 8.2 presents several values of the PCR C_p along with the associated values of process fallout, expressed in defective parts or nonconforming units of product per million (ppm).
- To illustrate the use of Table 8.2, notice that a PCR for a normally distributed stable process of $C_p = 1$ implies a fallout rate of 2,700 ppm for two sided specifications, whereas a PCR of $C_p = 1.5$ for this process implies a fallout rate of 4 ppm for one-sided specifications.

■ TABLE 8.2

Values of the Process Capability Ratio (C_p) and Associated Process Fallout for a Normally Distributed Process (in Defective ppm) That Is in Statistical Control

PCR	Process Fallout (in defective ppm)	
	One-Sided Specifications	Two-Sided Specifications
0.25	226,628	453,255
0.50	66,807	133,614
0.60	35,931	71,861
0.70	17,865	35,729
0.80	8,198	16,395
0.90	3,467	6,934
1.00	1,350	2,700
1.10	484	967
1.20	159	318
1.30	48	96
1.40	14	27
1.50	4	7
1.60	1	2
1.70	0.17	0.34
1.80	0.03	0.06
2.00	0.0009	0.0018

Assumptions for Interpretation of Numbers in Table 8.2

- The ppm quantities in Table 8.2 were calculated using the following important assumptions:

1. The quality characteristic has a normal distribution.
2. The process is in statistical control.
3. In the case of two-sided specifications, the process mean is centered between the lower and upper specification limits.

- Violation of these assumptions can lead to big trouble in using the data in Table 8.2.

Some Recommended Guidelines for Minimum Values of the PCR

Recommended Minimum Values of the Process Capability Ratio

	Two-Sided Specifications	One-Sided Specifications
Existing processes	1.33	1.25
New processes	1.50	1.45
Safety, strength, or critical parameter, existing process	1.50	1.45
Safety, strength, or critical parameter, new process	1.67	1.60

A Measure of **Actual** Capability

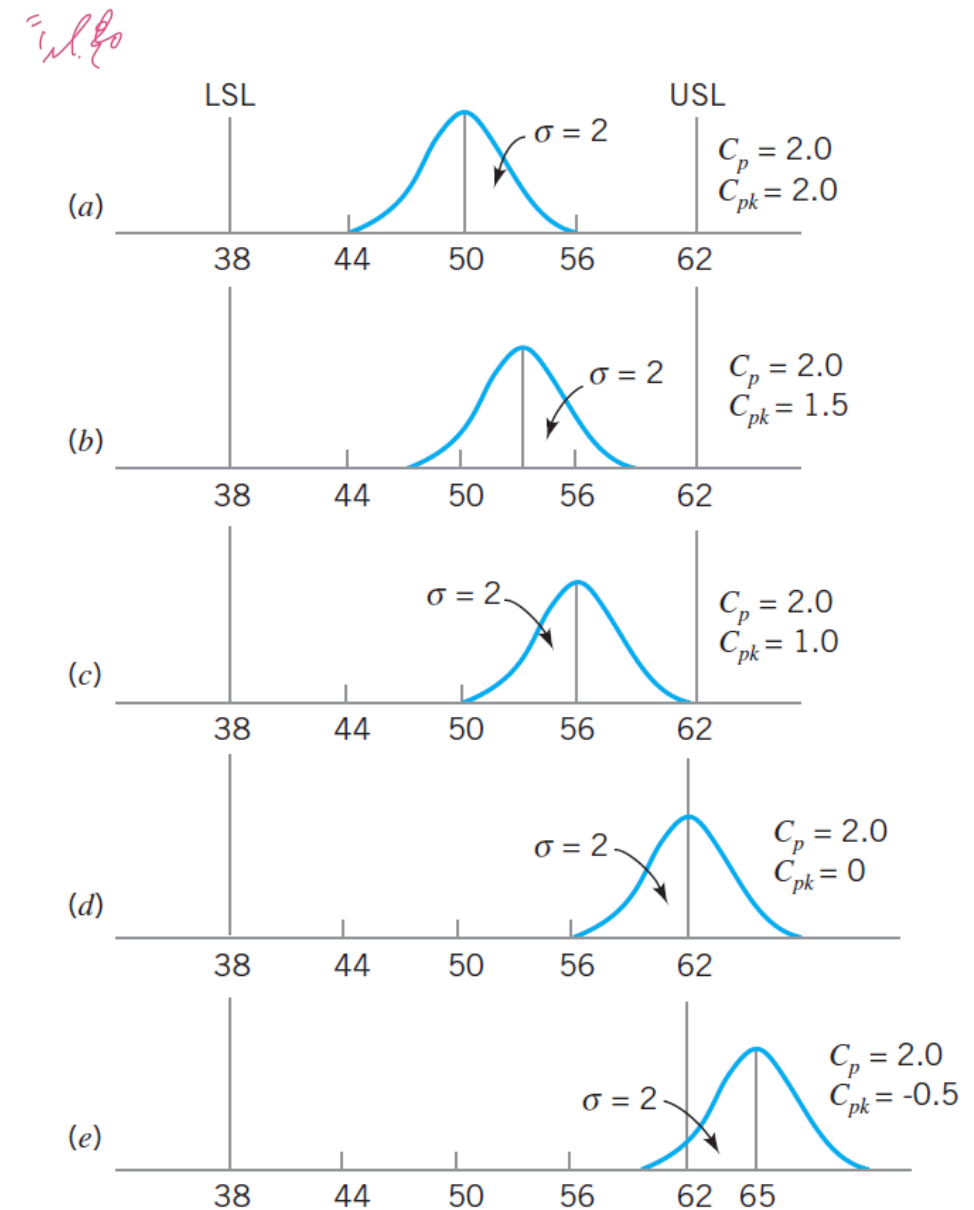
- C_p does not take process centering into account
- It is a measure of **potential** capability, not **actual** capability
- Thus, defining a new process capability ratio (PCR)— C_{pk} —that takes process centering into account and measure actual capability is needed.
- This quantity is


$$C_{pk} = \min(C_{pu}, C_{pl})$$


- For example, the normal distributions in Figure 8.8 have $C_p = 2$, but the process:

- in panel (b) of the figure clearly has lower capability than the process in panel (a) because it is not operating at the midpoint of the interval between the specifications.
- in panel (c), (d) and (e) $C_{pk} < C_p$, which means the process is off center.

- In general, if $C_{pk} = C_p$, the process is centered at the midpoint of the specifications, and when $C_{pk} < C_p$ the process is off center.



■ FIGURE 8.8 Relationship of C_p and C_{pk} .

Normality and Process Capability Ratios

- The assumption of normality is critical to the usual interpretation of these ratios (such as Table 8.2)
- For non-normal data, options are
 1. Transform non-normal data to normal
 2. Extend the usual definitions of PCR's to handle non-normal data
 3. Modify the definitions of PCR's for general families of distributions

Other Types of Process Capability Ratios

- First generation C_p
- Second generation C_{pk} and $C_{pm} = \frac{C_p}{\sqrt{1 + \left(\frac{\mu - 0.5(USL + LSL)}{\sigma}\right)^2}}$
- Third generation $C_{pkm} = \frac{C_{pk}}{\sqrt{1 + \left(\frac{\mu - 0.5(USL + LSL)}{\sigma}\right)^2}}$
- Lots of research has been done to develop ratios that overcome some of the problems with the basic ones
- Not much evidence that these ratios are used to any significant extent in practice

Process Capability Analysis Using a Control Chart

- Histograms, probability plots, and process capability ratios summarize the performance of the process. They do not necessarily display the potential capability of the process because they do not address the issue of statistical control, or show systematic patterns in process output that, if eliminated, would reduce the variability in the quality characteristic.
- Control charts are very effective in this regard. The control chart should be regarded as the primary technique of process capability analysis.
- Both attributes and variables control charts can be used in process capability analysis.

- Table 8.5 presents the container bursting-strength data in 20 samples of five observations each. The calculations for the $\bar{x}$ and R charts are summarized here:

$\bar{x}$ Chart

Center line = $\bar{\bar{x}} = 264.06$

$$UCL = \bar{\bar{x}} + A_2 \bar{R} = 264.06 + (0.577)(77.3) = 308.66$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R} = 264.06 - (0.577)(77.3) = 219.46$$

R Chart

Center line = $\bar{R} = 77.3$

$$UCL = D_4 \bar{R} = (2.115)(77.3) = 163.49$$

$$LCL = D_3 \bar{R} = (0)(77.3) = 0$$

TABLE 8.5

Glass Container Strength Data (psi)

Sample	Data					$\bar{x}$	R
1	265	205	263	307	220	252.0	102
2	268	260	234	299	215	255.2	84
3	197	286	274	243	231	246.2	89
4	267	281	265	214	318	269.0	104
5	346	317	242	258	276	287.8	104
6	300	208	187	264	271	246.0	113
7	280	242	260	321	228	266.2	93
8	250	299	258	267	293	273.4	49
9	265	254	281	294	223	263.4	71
10	260	308	235	283	277	272.6	73
11	200	235	246	328	296	261.0	128
12	276	264	269	235	290	266.8	55
13	221	176	248	263	231	227.8	87
14	334	280	265	272	283	286.8	69
15	265	262	271	245	301	268.8	56
16	280	274	253	287	258	270.4	34
17	261	248	260	274	337	276.0	89
18	250	278	254	274	275	266.2	28
19	278	250	265	270	298	272.2	48
20	257	210	280	269	251	253.4	70
						$\bar{\bar{x}} = 264.06$	$\bar{R} = 77.3$

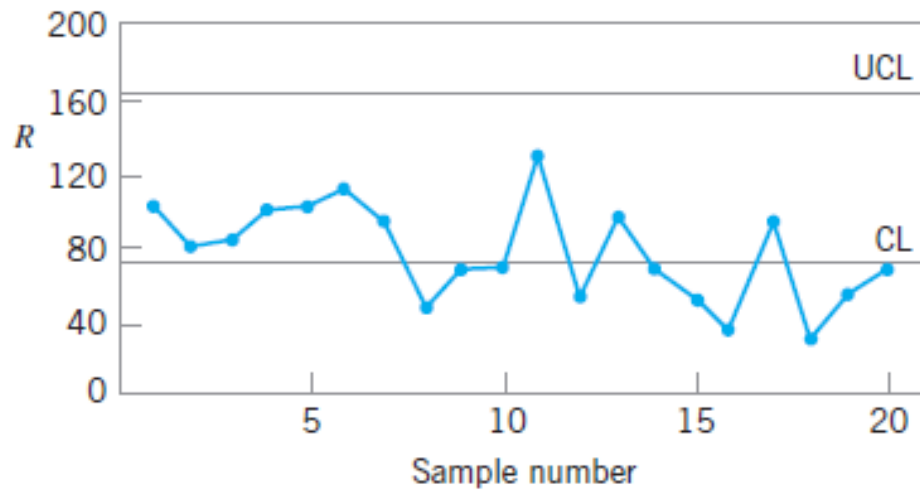
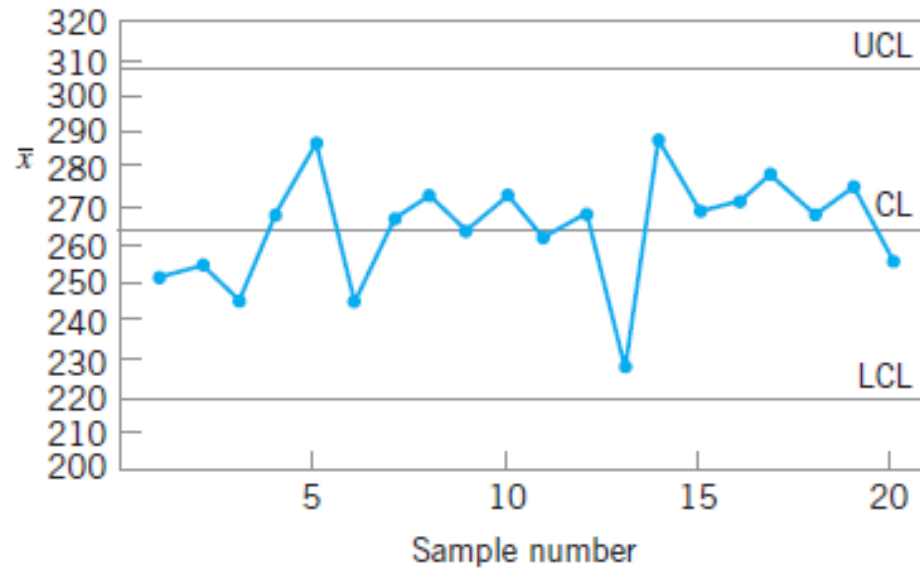


FIGURE 8.12 $\bar{x}$ and R charts for the bottle-strength data.

$$\hat{\mu} = \bar{\bar{x}} = 264.06$$

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{77.3}{2.326} = 33.23$$

Since $LSL = 200$

$$\hat{C}_{pl} = \frac{\mu - LSL}{3\hat{\sigma}} = \frac{264.06 - 200}{3(33.23)} = 0.64$$

8.7 Gauge and Measurement Systems Capability Studies

- Determining the capability of the measurement system is an important aspect of many quality and process improvement activities.
- Generally, in any activity involving measurements, some of the observed variability will be inherent in the units or items that are being measured, and some of the variability will result from the measurement system that is used.
- The measurement system will consist (minimally) of an instrument or gauge, and it often has other components, such as the operator(s) that uses it and the conditions or different points in time under which the instrument is used.
- There may also be other factors that impact measurement system performance, such as setup or calibration activities.

8.7 Gauge and Measurement Systems Capability Studies

- The purpose of most measurement systems capability studies is to:
 1. Determine how much of the observed variability is due to the gauge or measurement system
 2. Isolate the components of variability in the measurement system
 3. Assess whether the gauge is capable (suitable for the intended application)

8.7 Gauge and Measurement Systems Capability Studies

- There are two R's of measurement systems capability:
 1. **repeatability** (Do we get the same observed value if we measure the same unit several times under identical conditions?), and
 2. **reproducibility** (How much difference in observed values do we experience when units are measured under different conditions, such as different operators, time periods, and so forth?).
- Other important aspects of measurement systems capability: **linearity** (result of calibration and maintenance issues), **stability** (result from warm-up effects, environmental factors, inconsistent operator performance, and inadequate standard) and **bias** (reflects the difference between observed measurements and a “true” value obtained from a master or gold standard).

8.7 Gauge and Measurement Systems Capability Studies

- To introduce some of the basic ideas of measurement systems analysis (MSA), consider a simple but reasonable model for measurement system capability studies

$$y = x + \varepsilon \quad (8.23)$$

where y is the total observed measurement, x is the true value of the measurement on a unit of product, and ε is the measurement error.

- Assume that x and ε are normally and independently distributed random variables with means μ and 0 and variances (σ_P^2) and (σ_{Gauge}^2) , respectively. The variance of the total observed measurement, y , is then

$$\sigma_{Total}^2 = \sigma_P^2 + \sigma_{Gauge}^2$$

EXAMPLE 8.7 Measuring Gauge Capability

An instrument is to be used as part of a proposed SPC implementation. The quality-improvement team involved in designing the SPC system would like to get an assessment of gauge capability. Twenty units of the product are obtained, and the process operator who will actually take the measurements for the control chart uses the instrument to measure each unit of product twice. The data are shown in Table 8.6.

■ TABLE 8.6
Parts Measurement Data

Part Number	Measurements		$\bar{x}$	R
	1	2		
1	21	20	20.5	1
2	24	23	23.5	1
3	20	21	20.5	1
4	27	27	27.0	0
5	19	18	18.5	1
6	23	21	22.0	2
7	22	21	21.5	1
8	19	17	18.0	2
9	24	23	23.5	1
10	25	23	24.0	2
11	21	20	20.5	1
12	18	19	18.5	1
13	23	25	24.0	2
14	24	24	24.0	0
15	29	30	29.5	1
16	26	26	26.0	0
17	20	20	20.0	0
18	19	21	20.0	2
19	25	26	25.5	1
20	19	19	19.0	0

- Figure 8.14 shows the $\bar{x}$ and R charts for these data. Note that **the chart exhibits many out-of-control points**. This is because the chart in this example shows the discriminating power of the instrument—literally, **the ability of the gauge to distinguish between units of product**.
- The R chart directly shows the magnitude of measurement error, or the gauge capability.
- The R values represent the difference between measurements made on the same unit using the same instrument.
- In this example, the R chart is in control. This indicates that the operator is having no difficulty in making consistent measurements. Out-of-control points on the R chart could indicate that the operator is having difficulty using the instrument.
- The standard deviation of measurement error, σ_{Gauge} , can be estimated as follows:

$$\hat{\sigma}_{Gauge} = \frac{\bar{R}}{d_2} = \frac{1}{1.128} = 0.887$$

- The distribution of measurement error is usually well approximated by the normal. Thus, $\hat{\sigma}_{Gauge}$ is a good estimate of gauge capability.

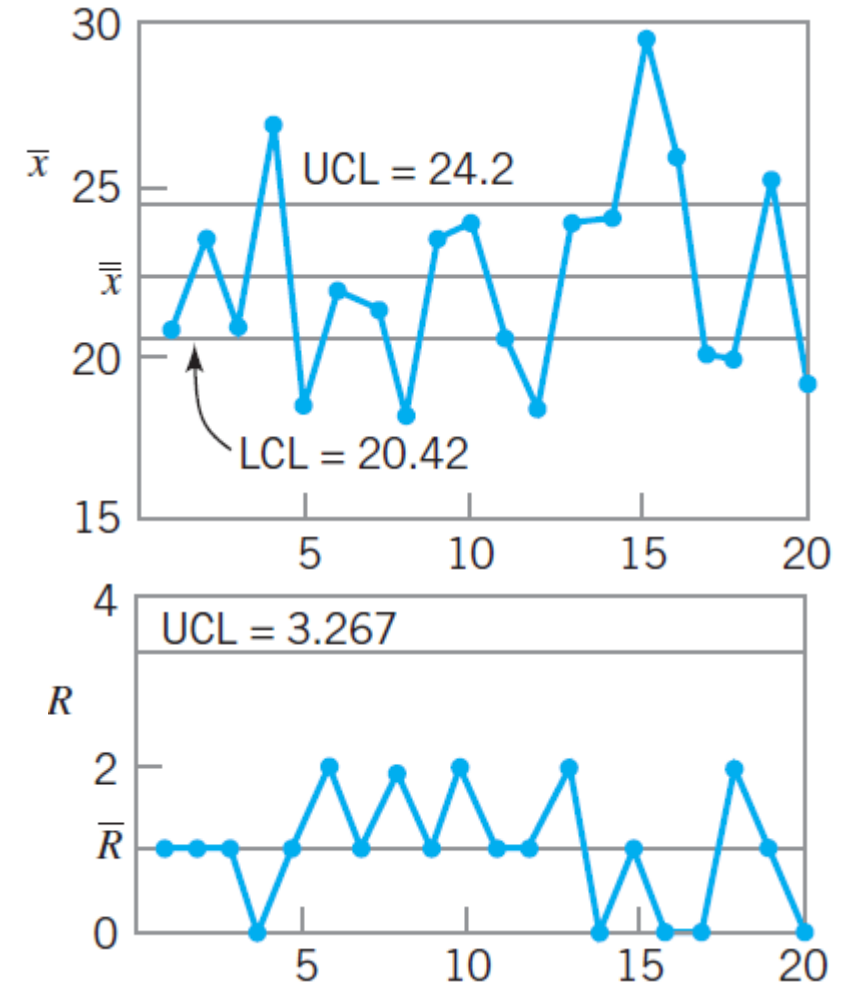


FIGURE 8.14 Control charts for the gauge capability analysis in Example 8.7.

The P/T ratio

- It is a fairly common (but not necessarily good) practice to compare the estimate of gauge capability to the width of the specifications or the tolerance band ($USL - LSL$) for the part that is being measured.
- The ratio of to the total tolerance band is often called the **precision-to-tolerance (P/T) ratio**:

$$* \quad P/T = \frac{k \hat{\sigma}_{Gauge}}{USL - LSL}$$

- Popular choices for the constant k are $k = 5.15$ and $k = 6$.
- The value $k = 5.15$ corresponds to the limiting value of the number of standard deviations between bounds of a 95% tolerance interval that contains at least 99% of a normal population, and $k = 6$ corresponds to the number of standard deviations between the usual natural tolerance limits of a normal population.

The P/T ratio

- For $\hat{\sigma}_{Guage} = 0.887$ and $USL = 60$, $LSL = 5$ and $k = 6$, then the estimate of the P/T ratio is:

$$P/T = \frac{6 (0.887)}{60 - 5} = \frac{5.32}{55} = 0.097$$

- Values of the estimated ratio P/T of 0.1 or less often are taken to imply adequate gauge capability.

 - This is based on the generally used rule that requires a measurement device to be calibrated in units one-tenth as large as the accuracy required in the final measurement. However, we should use caution in accepting this general rule of thumb in all cases. A gauge must be sufficiently capable of measuring product accurately enough and precisely enough so that the analyst can make the correct decision. This may not necessarily require that $P/T \leq 0.1$.

Estimating the Variance Components

■ TABLE 8.6

Parts Measurement

Part Number	Measurements		$\bar{x}$	R
	1	2		
1	21	20	20.5	1
2	24	23	23.5	1
3	20	21	20.5	1
4	27	27	27.0	0
5	19	18	18.5	1
6	23	21	22.0	2
7	22	21	21.5	1
8	19	17	18.0	2
9	24	23	23.5	1
10	25	23	24.0	2
11	21	20	20.5	1
12	18	19	18.5	1
13	23	25	24.0	2
14	24	24	24.0	0
15	29	30	29.5	1
16	26	26	26.0	0
17	20	20	20.0	0
18	19	21	20.0	2
19	25	26	25.5	1
20	19	19	19.0	0

$$s = \sqrt{\frac{\sum_{i=1, j=1}^{mn} (x_{ij} - \bar{\bar{x}})^2}{mn - 1}} = \sqrt{\frac{\sum_{i=1, j=1}^{mn} (x_{ij} - 22.3)^2}{20 \times 2 - 1}} = \sqrt{\frac{392.4}{39}} = 3.17$$

This is an estimate of the standard deviation of **total variability**, including both **product variability** and **gauge variability**.

Therefore,

$$\hat{\sigma}_{Total}^2 = s^2 = (3.17)^2 = 10.05$$

Since

$$\sigma_{Total}^2 = \sigma_P^2 + \sigma_{Gauge}^2$$

we can obtain an estimate of $\hat{\sigma}_P^2$ as

$$\hat{\sigma}_P^2 = \hat{\sigma}_{Total}^2 - \hat{\sigma}_{Gauge}^2 = (3.17)^2 - (0.887)^2 = 9.26$$

Therefore, an estimate of the standard deviation of the product characteristic is

$$\hat{\sigma}_P = \sqrt{9.26} = 3.04$$

There are other measures of gauge capability that have been proposed. One of these is the ratio of process (part) variability to total variability:

$$\rho_P = \frac{\sigma_P^2}{\sigma_{\text{Total}}^2} \quad (8.26)$$

and another is the ratio of measurement system variability to total variability:

$$\rho_M = \frac{\sigma_{\text{Gauge}}^2}{\sigma_{\text{Total}}^2} \quad (8.27)$$

Obviously, $\rho_P = 1 - \rho_M$. For the situation in Example 8.7 we can calculate an estimate of ρ_M as follows:

$$\hat{\rho}_M = \frac{\hat{\sigma}_{\text{Gauge}}^2}{\hat{\sigma}_{\text{Total}}^2} = \frac{0.79}{10.05} = 0.0786 \quad \approx 0.1$$

Thus the variance of the measuring instrument contributes about 7.86% of the total observed variance of the measurements.

Another measure of measurement system adequacy is defined by the AIAG (1995) [note that there is also an updated edition of this manual, AIAG (2002)] as the signal-to-noise ratio (SNR):

$$\textcircled{8} \quad SNR = \sqrt{\frac{2\rho_P}{1 - \rho_P}} \quad (8.28)$$

AIAG defined the SNR as the number of distinct levels or categories that can be reliably obtained from the measurements. A value of 5 or greater is recommended, and a value of less than 2 indicates inadequate gauge capability. For Example 8.7 we have $\hat{\rho}_M = 0.0786$, and using $\hat{\rho}_P = 1 - \hat{\rho}_M$ we find that $\hat{\rho}_P = 1 - \hat{\rho}_M = 1 - 0.0786 = 0.9214$, so an estimate of the SNR in equation 8.28 is

$$\textcircled{9} \quad SNR = \sqrt{\frac{2\hat{\rho}_P}{1 - \hat{\rho}_P}} = \sqrt{\frac{2(0.9214)}{1 - 0.9214}} = 4.84$$

Therefore, the gauge in Example 8.7 would not meet the suggested requirement of an SNR of at least 5. However, this requirement on the SNR is somewhat arbitrary.

Discrimination Ratio

Another measure of gauge capability is the discrimination ratio (DR)

$$DR = \frac{1 + \rho_P}{1 - \rho_P} \quad DR > 4 \quad (8.29)$$

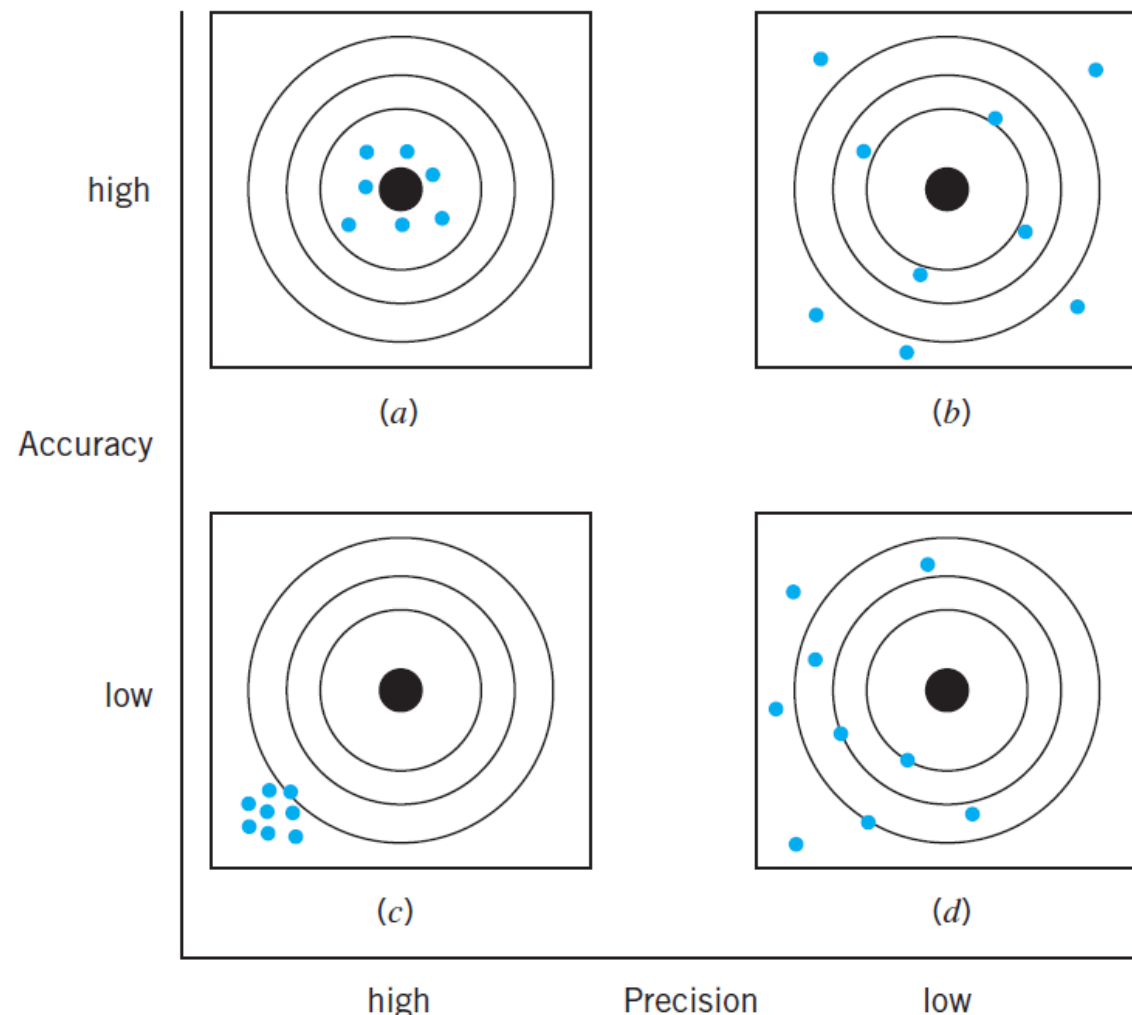
Some authors have suggested that for a gauge to be capable the DR must exceed 4. This is a very arbitrary requirement. For the situation in Example 8.7, we would calculate an estimate of the discrimination ratio as

$$\hat{DR} = \frac{1 + \rho_P}{1 - \rho_P} = \frac{1 + 0.9214}{1 - 0.9214} = 24.45$$

Clearly by this measure, the gauge is capable.

Accuracy and Precision

- We have focused primarily on instrument or **gauge precision**, not **gauge accuracy**.
- These two concepts are illustrated in Figure 8.15. In this figure, the bull's-eye of the target is considered to be the true value of the measured characteristic, or μ the mean of x in equation 8.23.
- Accuracy refers to the ability of the instrument to measure the true value correctly on average, whereas precision is a measure of the inherent variability in the measurement system.
- Evaluating the accuracy of a gauge or measurement system often requires the use of a standard, for which the true value of the measured characteristic is known. Often the accuracy feature of an instrument can be modified by making adjustments to the instrument or by the use of a properly constructed calibration curve.



■ **FIGURE 8.15** The concepts of accuracy and precision. (a) The gauge is accurate and precise. (b) The gauge is accurate but not precise. (c) The gauge is not accurate but it is precise. (d) The gauge is neither accurate nor precise.

- It is also possible to design measurement systems capability studies to investigate **two components of measurement error**, commonly called the **repeatability and the reproducibility of the gauge**.
- We define **reproducibility** as the variability due to different operators using the gauge (or different time periods, or different environments, or in general, different conditions) and **repeatability** as reflecting the basic inherent precision of the gauge itself.
- That is,

$$\star \quad \sigma_{\text{Measurement Error}}^2 = \sigma_{\text{Gauge}}^2 = \sigma_{\text{Repeatability}}^2 + \sigma_{\text{Reproducibility}}^2 \quad (8.30)$$

- The experiment used to measure the components of σ_{Gauge}^2 is usually called a gauge R & R study, for the two components of σ_{Gauge}^2 .

Important Terms and Concepts

ANOVA approach to a gauge R & R experiment
Components of gauge error
Components of measurement error
Confidence intervals for gauge R & R studies

Discrimination ratio (DR) for a gauge
Estimating variance components
Factorial experiment
Gauge R & R experiment
Graphical methods for process capability analysis
Measurement systems capability analysis
Natural tolerance limits for a normal distribution
Natural tolerance limits of a process
Nonparametric tolerance limits
Normal distribution and process capability ratios
One-sided process-capability ratios
PCR C_p
PCR C_{pk}

Confidence intervals on process capability ratios
Consumer's risk or missed fault for a gauge
Control charts and process capability analysis
Delta method

PCR C_{pm}
Precision and accuracy of a gauge
Precision-to-tolerance (P/T) ratio
Process capability
Process capability analysis
Process performance indices P_p and P_{pk}
Producer's risk or false failure for a gauge
Product characterization
Random effects model ANOVA
Signal-to-noise ratio (SNR) for a gauge
Tolerance stack-up problems
Transmission of error formula

Learning Objectives

1. Investigate and analyze process capability using control charts, histograms, and probability plots
2. Understand the difference between process capability and process potential
3. Calculate and properly interpret process capability ratios
4. Understand the role of the normal distribution in interpreting most process capability ratios
5. Calculate confidence intervals on process capability ratios
6. Know how to conduct and analyze a measurement systems capability (or gauge R & R) experiment
7. Know how to estimate the components of variability in a measurement system
8. Know how to set specifications on components in a system involving interaction components to ensure that overall system requirements are met
9. Estimate the natural limits of a process from a sample of data from that process